Syrian Arab Republic

Syrian Private University

Faculty of Computer Engineering

and Information Technology

Specialization: Artificial Intelligence Engineering

semester project

Study of Human Activity (Sports Activity)

Prepared by:

Abdulkarim Samra

Supervised by: Dr. Majda Al-Bakour

May 2026

Supervisor's Certificate:

Name:...................................

Date:.....................................

Signature:..............................

Abstract:

The study of human activity, particularly athletic activity, is a crucial research field that integrates health science, technology, and data science. It plays a fundamental role in understanding human motor behavior and analyzing physical performance. This type of research aims to analyze the physical movements performed by individuals during various sports activities, such as walking, running, jumping, and fitness exercises, to improve athletic performance, reduce the risk of injury, and promote overall health.

The study of athletic activity relies on collecting kinetic data using modern technologies such as wearable sensors, inertial measurement units, and smartphones. These technologies allow for the precise recording of information on acceleration, speed, direction, and rate of movement. After data collection, it is processed and analyzed using advanced algorithms and machine learning techniques to extract the distinctive movement patterns of each sport, enabling high-precision differentiation between various activities.

The results of these studies contribute to several fields, most notably the development of customized training programs for athletes, improving the efficiency of athletic exercises, and objectively monitoring physical progress. They are also used in the medical field for motor rehabilitation, helping to monitor patients' conditions and assess their motor development after injuries or surgeries. Furthermore, it plays a crucial role in smart health monitoring systems designed to promote an active lifestyle and reduce physical inactivity.

The study of physical activity represents a bridge between humans and technology, contributing to the development of intelligent systems capable of understanding and responding to human movement. Through continuous advancements in sensing and analysis technologies, access to these systems has become increasingly possible

To more accurate and reliable results, making this field a cornerstone in the development of modern health and sports applications, and a major supporter of future research in the field of human activity.

Table of Contents:

Chapter One:

Chapter Two:

Chapter Three:

Table Index:

Figure Index:

Glossary Table:

Arabic Name	English Name	Explanation
Artificial Intelligence	Artificial Intelligence	A branch of computer science aimed at designing systems capable of simulating human intelligence, such as learning, reasoning, and decision-making.
Machine Learning	Machine Learning	A subfield of artificial intelligence that relies on algorithms to learn patterns from data without being explicitly programmed.
Deep Learning	Deep Learning	A subfield of machine learning that utilizes deep neural networks to extract complex representations from data.
Computer Vision	Computer Vision	A field focused on enabling computers to understand and analyze digital images and videos.
Human Pose Estimation	Human Pose Estimation	A technique aimed at identifying the locations of human body joints in images or videos using deep learning models.
MediaPipe	MediaPipe	An open-source framework developed by Google used for real-time body landmark extraction and tracking.
Landmark	Landmark	A point representing a specific body joint, such as the shoulder, knee, or elbow.
Coordinates	Coordinates	Numerical values expressing the position of a landmark in an image using the x and y axes.
Features	Features	Numerical values extracted from data used as inputs for model training.

Arabic Name	**English Name**	**Explanation**
Joint Angles	Joint Angles	Angles resulting from three landmarks representing a specific joint, expressing the movement of that body part.
Normalized Distances	Normalized Distances	Distances between body landmarks after being normalized relative to a reference distance, to reduce the impact of varying body sizes.
Normalization	Normalization	A process aimed at standardizing the range of numerical feature values to improve model performance.
Temporal Sequence	Temporal Sequence	A consecutive sequence of frames representing body movement over time.
Sequence Length	Sequence Length	The number of frames used to represent a movement, which has been set to 30 frames in the project.
Recurrent Neural Networks	Recurrent Neural Networks	A type of neural network designed to process sequential data, such as time series.

Arabic Name	**English Name**	**Explanation**
Recurrent Neural Networks	Recurrent Neural Networks	A type of neural network designed to process sequential data, such as time series.
Bidirectional Long Short-Term Memory (BiLSTM)	Bidirectional Long Short-Term Memory (BiLSTM)	A bidirectional recurrent neural network capable of learning temporal patterns from both the past and the future simultaneously.
Training	Training	The phase where the model learns from the training data with the goal of minimizing error.
Validation	Validation	The phase of evaluating the model's performance during training using data that was not used in the learning process.
Testing	Testing	The phase of evaluating the final model using new, unseen data that it has never encountered before.
Confusion Matrix	Confusion Matrix	A statistical tool used to analyze the performance of a classification model.
Accuracy	Accuracy	The ratio of correct predictions to the total number of predictions made.
Recall	Recall	The model's ability to correctly detect positive instances for each class.
Precision	Precision	The ratio of correctly predicted positive observations to the total predicted positives.
F1-Score	F1-Score	A metric that balances both precision and recall to evaluate the overall performance of the model.

Arabic Name	**English Name**	**Explanation**
Label Encoding	Label Encoding	Converting textual labels into numerical values so that the model can process them.
Label Decoding	Label Decoding	Converting the numerical values outputted by the model back into the original exercise names.

Chapter 1:

Project Introduction

1.1 Introduction:

The world is currently witnessing rapid advancements in digital technology and artificial intelligence, which have contributed to the application of these technologies in the study and analysis of human behavior in its various forms. Physical activity is one of the most prominent manifestations of this behavior. This project focuses on studying human activity related to physical movement and sports activities by analyzing kinematic data to understand movement patterns, evaluate physical performance, and improve quality of life and overall health.

The importance of this project stems from the growing need for intelligent systems capable of accurately and objectively monitoring and analyzing physical activity, whether for ordinary individuals or professional athletes. Studying physical activity allows for a comprehensive assessment of physical activity levels, early detection of movement errors, and a contribution to reducing injuries resulting from incorrect sports practices. These studies also play a crucial role in supporting sports training and rehabilitation programs.

The project relies on the use of modern data collection technologies, such as wearable sensors or smart devices, which record various movement signals during sports activities. This data is then processed and analyzed using algorithms and machine learning techniques to extract the distinctive characteristics of each activity and build models capable of accurately identifying and classifying sports activities. This project seeks to integrate theory and practice by presenting a practical model demonstrating how to utilize kinetic data in building an intelligent system for studying athletic activity. It also aims to highlight the role of modern technology in

Serving the health and sports sector, this project opens new horizons for developing smart applications that contribute to improving physical performance, promoting an active lifestyle, and supporting future research in the field of human activity studies.

2.1 – Project Methodology:

The methodology of this project relies on following a series of integrated systematic steps to analyze and classify human athletic activity using computer vision and deep learning techniques, starting with data collection and ending with evaluating the performance of the proposed model.

In the first phase, the project's objective was clearly defined: to build an intelligent system capable of recognizing different athletic exercises based on body movement patterns extracted from video clips.

In the second phase, data was collected in the form of video clips representing a range of different athletic activities. Human movements were recorded using a standard camera, eliminating the need for any wearable sensors, which simplifies the system and increases its practicality.

This was followed by a data preprocessing phase, where the MediaPipe framework was used to extract human joint points from each video frame. Next, this data was organized and processed by removing invalid frames and standardizing the input format, dividing the data into fixed-length time sequences of 30 frames each to dynamically represent the movement.

In the following stage, the most expressive kinematic characteristics of the exercises were extracted, including the angles of the major joints and the distances.

The standardization of body parts, normalized using a reference distance, aims to minimize the impact of varying object sizes and distances from the camera.

These characteristics are key inputs for the proposed model due to their high capacity for accurately representing movement.

Next, deep learning algorithms, specifically a Bidirectional Long-Term Memory Network (LSTM), were used to learn and classify the temporal patterns of athletic movements.

The model was trained using training data and then evaluated using an independent validation set, relying on standard performance metrics such as accuracy, recall, and F1 score.

In the final stage, the model's results were analyzed and presented using scientific evaluation tools such as the ambiguity matrix and classification reports. The model was also tested on new video clips to verify its generalizability.

The results demonstrated the effectiveness of the methodology in achieving the project objectives, confirming the potential of this approach in developing intelligent systems for analyzing human athletic activity.

1.2.1 – Project Research Problem:

Despite significant advancements in computer vision and machine learning, the study and analysis of human athletic activity still faces numerous research and application challenges. The research problem for this project lies in the difficulty of accurately and reliably identifying and automatically analyzing various athletic activities using video footage, especially given the diversity of movement patterns among individuals, the differences in physical performance styles, and the varying filming conditions such as camera angle, lighting, and distance from the user.

Many traditional methods for evaluating athletic performance rely on direct observation or manual assessment by coaches. These methods are often subject to subjective error and lack accuracy and consistency.

Furthermore, some intelligent video analysis systems suffer from limitations in their ability to handle visual noise, the loss of some body joints during movement, and the instability of motion point detection in certain frames, negatively impacting the accuracy of the analysis results.

Additionally, a significant challenge arises in the difficulty of extracting the most expressive movement characteristics of athletic exercises, particularly when there are similarities in movement patterns between certain exercises, such as arm exercises or lower body exercises. Choosing the right algorithms that are capable of accurately modeling the timeline of movement and distinguishing between these activities is also one of the main challenges in this field.

The research problem for this project stems from the need to design an intelligent and efficient system that uses video analysis to extract and process kinematic characteristics, with the aim of reliably identifying and classifying exercises.

This project seeks to address these challenges by employing computer vision and deep learning technologies, thus contributing to the development of modern applications in the fields of sports training and digital health.

2.2.1 - Project Objectives:

This project aims to design and develop an intelligent system for analyzing human athletic activity using computer vision and deep learning technologies. This involves processing video clips and extracting the kinematic characteristics that represent the movement of the human body during exercise.

The project aims to provide a practical framework capable of accurately identifying the type of exercise by relying on temporal analysis of the movement sequence, rather than relying on traditional human assessment or manual measurements.

The detailed objectives of the project are as follows:

1- To build a system capable of automatically identifying a range of different exercises, such as arm exercises, push-ups, and squats, based on the analysis of movement extracted from video clips.

2- Utilizing pose estimation techniques to extract the human body's articulation points from each video frame, enabling the representation of human movement in a digital, organized, and computationally manageable manner.

3. Converting visual (video) data into numerical data representing joint coordinates and angles between joints, enabling its use as effective input for machine learning and deep learning models.

4. Processing the extracted kinematic data by filtering out noise and addressing loss of detection at certain joint points, aiming to improve data quality and enhance model performance stability.

5. Extracting unique kinematic characteristics for each exercise, such as angles between joints and standard relative distances, which helps mitigate the impact of differences in body height or camera angle among users.

6. Dividing the kinematic data into fixed-length time sequences (sequences) representing successive phases of movement, allowing the model to

understand the temporal structure of the exercise rather than simply analyzing separate frames.

7. Designing and training a model based on Bidirectional Recurrent Neural Networks (BiLSTM) to utilize preceding and subsequent time information in the movement sequence, thus enhancing the model's ability to distinguish between exercises that are similar in form but differ in their movement sequence. 8. Evaluate the performance of the proposed model using a set of established scientific metrics, such as accuracy, confusion matrix, precision, recall, and F1-score, to comprehensively measure the model's efficiency and the reliability of its results.

9. Study the system's generalizability when applied to new, unused training videos to verify the model's ability to handle user variations and different recording conditions.

10. Contribute to supporting smart sports and health applications by providing an analytical system that can be used in the future to evaluate athletic performance, improve training quality, and track physical activity objectively and automatically.

3.2.1 – Background Study:

The field of Human Activity Recognition (HARR) has witnessed remarkable development in recent years, driven by significant advancements in sensing technologies, computer vision, and machine learning and deep learning algorithms.

Many previous studies have focused on developing intelligent systems capable of accurately recognizing athletic activities and analyzing human movement using various data sources and methods.

Some studies have relied on wearable sensors, such as accelerometers and gyroscopes, due to their ability to provide accurate motion data.

However, this type of system requires additional equipment and direct user interaction, which may limit its ease of use in practical applications.

In contrast, other studies have utilized video footage and computer vision technologies to extract body movement without the need for additional devices.

With the emergence of pose estimation technologies, such as OpenPose and MediaPipe, it has become possible to extract the human body's articulated points and convert movement into analyzable digital data. Several studies have used these techniques to extract kinetic characteristics and then applied machine learning algorithms or recurrent neural networks (RNN and LSTM) to analyze the chronological sequence of movement and classify different activities.

Despite the promising results achieved in these studies, some challenges remain, such as the influence of camera angles, individual user differences, and the need for models capable of more accurately understanding the temporal structure of movement.

This project builds upon these previous works, focusing on using video analysis and deep learning techniques to improve the accuracy of exercise recognition.

A comparison of previous studies reveals a widespread interest in using posture estimation and motion analysis techniques to infer different activities from video or sensor data. Some studies rely on data enhancement techniques or integrate CNN models with LSTM to achieve better performance, while others utilize advanced networks like GCN to handle spatial relationships within the skeletal system. Although many of these studies achieve good results, challenges such as the influence of lighting, camera angle, and user differences persist, necessitating models capable of more accurately grasping the temporal dimension of movement. This is the focus of this project, which utilizes BiLSTM and standardized feature analysis.

Literature Review Comparison Table:

Study	Data Type Used	Main Technique	Algorithm Type	Features / Advantages	Source	Year
Augmenting Vision-Based Human Pose Estimation with Rotation Matrix	Video + Body Landmarks	Enhancing computer vision techniques	SVM with Data Augmentation	Achieved an accuracy of up to 96% using Augmentation	arXiv	2021
A Comprehensive Review on Human Activity and Fitness Tracker using Different Approaches	Video + Images + Motion Analysis	Comprehensive review of pose estimation techniques	Various techniques (OpenPose, CNN)	Provides a comparison between different computer vision models with an accuracy of up to 94%	IJRASET	2021
Human pose estimation and action recognition on fitness behavior and fitness	Video	Motion analysis networks (GCN)	Spatio-Temporal Graph Convolutional Networks	Demonstrates the impact of graph networks in action recognition with an accuracy of up to 95%	ScienceDirect	2022
Fitcam: detecting and counting repetitive exercises with deep learning	Video + Frame Analysis	Repetition detection and counting	LSTM and Binary Classification	An effective model for analyzing push-ups and squats, with LSTM evaluation reaching up to 94% accuracy	Springer Nature Link	2023

Study	Data Type Used	Main Technique	Algorithm Type	Features / Advantages	Source	Year
Combined CNN-LSTM for Recognizing Physical Activities	Sensor Data + Video	Hybrid CNN + LSTM combination	Hybrid Architecture	Integrates spatial and temporal information to achieve robust performance with up to 97% accuracy	Tech Science	2023
Skeleton-Based Action Recognition Using Spatio-Temporal LSTM	3D Skeleton Data	Skeleton Analysis + Temporal	Spatio-Temporal LSTM	Successfully handles noise in skeleton sequence data with an accuracy of 97%	arXiv	2024

Chapter Two:

Sports and Artificial Intelligence

1.2 - Introduction:

Sports are among the most important human activities that contribute to promoting physical and mental health. They also play a fundamental role in improving quality of life and preventing many chronic diseases. With the rapid development of digital technologies, the sports field no longer relies solely on human experience and traditional observation, but increasingly benefits from artificial intelligence and data analysis technologies.

Artificial intelligence contributes to the development of the sports field by analyzing physical movements with high accuracy, monitoring motor performance, and objectively evaluating the quality of athletic exercises. This relies on advanced technologies such as computer vision and deep learning, which enable the understanding of human motor activity from video clips and the extraction of important kinematic characteristics such as joint angles, relative distances between body parts, and the temporal sequence of movement.

Deep learning models, particularly recurrent neural networks such as LSTM and BiLSTM, have helped improve the accuracy of analyzing athletic activities due to their ability to represent the temporal relationships between successive movements. This is essential for understanding athletic exercises that rely on a sequence of movement positions. The use of posture estimation technologies, such as MediaPipe Pose, has also contributed to the ability to analyze movement based solely on video, eliminating the need for wearable sensors. This enhances the flexibility of smart systems and facilitates their application in diverse environments.

Based on this, this project aims to leverage artificial intelligence and deep learning technologies to identify exercises from short video clips by analyzing movement characteristics and chronological sequence. This approach offers broad possibilities for developing smart training applications, remote sports monitoring, and correction of movement errors, thereby supporting the digital transformation of the sports sector and improving the quality of sports and health services.

2.2 – Exercise Explanation and Types:

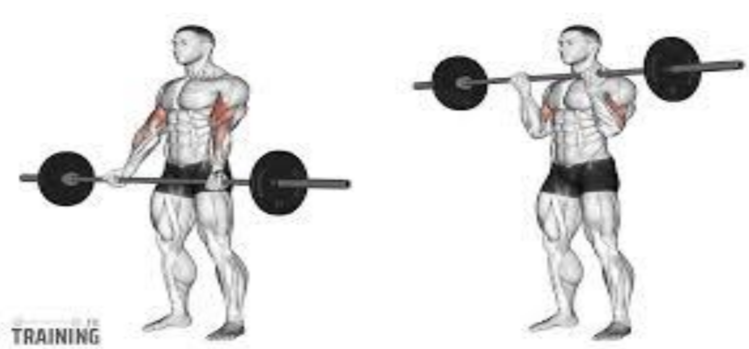
This project is based on the analysis of a group of common gym exercises, characterized by their diverse movement patterns and the variety of joints used in their execution. These exercises were carefully selected due to their suitability for video-based motion analysis techniques and their ease of differentiation based on joint angles and the chronological sequence of movement. A brief explanation of each exercise, along with its kinematic characteristics, follows.

1.2.2 Barbell Biceps Curl:

The Barbell Biceps Curl is one of the most popular arm strengthening exercises, primarily targeting the biceps. The exercise is performed by standing upright, holding the barbell with both hands, and then gradually lifting it from an extended arm position while bending the elbows. The curl is then slowly returned to the starting position.

This exercise relies mainly on elbow joint movement, with relative stability at the shoulder joint, making it suitable for analyzing joint angles in AI-powered human activity recognition systems. The repetitive movement pattern of the exercise also helps deep learning models to easily detect the chronological sequence of movement.

2.2.2



Hammer Curl:

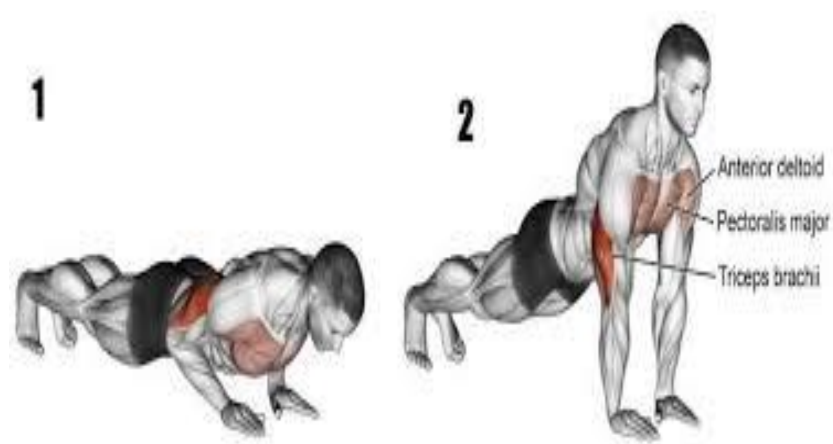
The Hammer Curl is similar to a traditional biceps curl, but it's performed with dumbbells facing inwards instead of horizontally. This exercise targets the brachialis and biceps muscles, and features more stable shoulder movement, with a clear emphasis on elbow flexion.

This difference in grip and movement direction is a key factor that helps AI systems distinguish between the Hammer Curl and the Barbell Biceps Curl, despite their general similarities, by analyzing joint angles and relative body positioning.

3.2.2 Push-Up Exercise:

The push-up is a fundamental bodyweight exercise that targets the chest, shoulders, arms, and core muscles. It is performed by lowering the body towards the ground and then pushing it back up using the arms, maintaining a straight line throughout the movement.

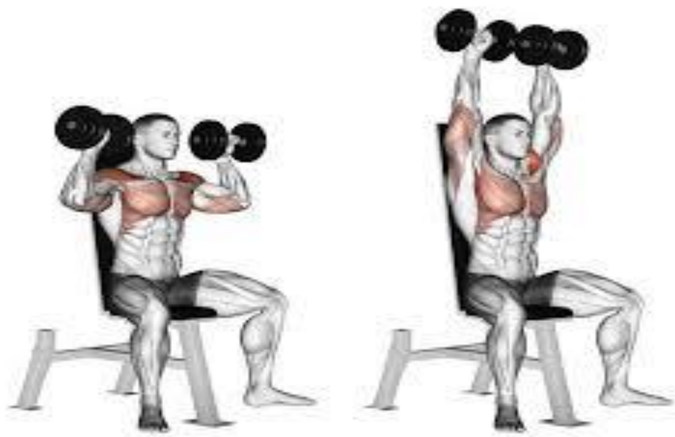
This exercise is characterized by simultaneous movement changes in several joints, such as the shoulder, elbow, and hip, making it rich in kinematic features. These features are well-suited for analyzing the temporal sequence of movement using repetitive neural networks, as the exercise requires continuous motor coordination between different body parts.



4.2.2 Shoulder Press Exercise:

The shoulder press is a fundamental exercise for strengthening the shoulder muscles. It is performed by lifting a weight from shoulder level to overhead using a vertical motion. This exercise relies on a significant change in the angles of the shoulder and elbow joints during the upward and downward movements.

These pronounced angular changes facilitate the recognition of the exercise within video-based motion analysis systems and allow for the extraction of distinctive kinematic characteristics that aid in its accurate classification using deep learning models.



5.2.2 Squat Exercise:

The squat is one of the most important exercises for strengthening the lower body, targeting the thigh, hip, and knee muscles. The exercise is performed by bending the knees and lowering the body, then returning to a standing position in a controlled manner.

This exercise is characterized by significant and noticeable changes in the angles of the knee and hip joints, making it easily detectable in kinematic analysis. Furthermore, the vertical movement of the body during the exercise provides a distinctive movement pattern that artificial intelligence systems can easily differentiate from other exercises.

3.2 Artificial Intelligence and Its Fundamental Concepts:

3.2.1 Definition of Intelligence:

Intelligence is defined as the ability to understand, learn, reason, and solve problems, as well as adapt to the surrounding environment and make appropriate decisions based on available data. It encompasses a range of cognitive processes, including analytical thinking, perceiving relationships, utilizing past experiences, and handling new situations with flexibility and efficiency.

In the technical field, the concept of intelligence is used to describe the capability of computer systems to simulate certain aspects of human intelligent behavior, such as learning from data, recognizing patterns, and making semi-autonomous decisions. This concept forms the foundation for modern artificial intelligence applications across various fields, including image, video, and motion analysis.

3.2.2 Machine Learning

Machine Learning is one of the core branches of artificial intelligence. It aims to enable computer systems to learn automatically from data without the need for explicit programming for every scenario. This field relies on building models capable of discovering patterns and relationships within data, and utilizing them for tasks such as prediction, classification, and decision-making.

Types of Machine Learning:

Machine learning is divided into several main types, most notably:

1. **Supervised Learning:** Relies on labeled data containing known inputs and outputs. It is extensively used in classification tasks, such as recognizing physical exercises.
2. **Unsupervised Learning:** Used to discover hidden patterns or groupings within unlabeled data.

3. **Semi-supervised Learning:** Combines a small amount of labeled data with a large amount of unlabeled data to improve learning accuracy.
4. **Reinforcement Learning:** Relies on the principle of reward and punishment to learn optimal decision-making policies.

General Architecture of Machine Learning

The architecture of machine learning systems typically consists of a training dataset, followed by a **Feature Extraction** phase. Next, a learning algorithm is applied to build a model capable of prediction or classification. Finally, the performance is evaluated using appropriate metrics.

3.2.3 Deep Learning:

Deep Learning is an advanced subfield of machine learning that relies on multi-layered artificial neural networks. This approach is uniquely characterized by its ability to automatically learn features directly from raw data without the need for manual feature engineering. Consequently, it is widely utilized in computer vision and video processing applications.

Types of Deep Learning Models:

Deep learning models include several key types, most notably **Convolutional Neural Networks (CNNs)**, which are used for image and video analysis, and **Recurrent Neural Networks (RNNs)**, designed for sequential data. Additionally, **LSTM** and **BiLSTM** networks are characterized by their ability to understand long-term temporal dependencies. Modern models also encompass **Generative Adversarial Networks (GANs)** and **Transformers**, which are utilized in advanced and multimodal applications.

General Architecture of Deep Learning

Deep networks consist of an input layer, multiple sequential hidden layers, and an output layer. In addition, they incorporate a loss function used to measure prediction error, and an optimization algorithm that updates the network's weights to minimize this error.

4.2 Neural Networks Used in the Project:

4.2.1 Convolutional Neural Networks (CNN):

Convolutional Neural Networks (CNNs) are among the most prominent deep learning models used in image and video analysis. These networks rely on convolution operations to automatically extract spatial features—such as edges, shapes, and visual patterns—directly from visual data.

The basic architecture of CNNs consists of:

1. **Convolution Layer:** Extracts spatial features such as edges and shapes.
2. **Activation Layer:** Introduces non-linearity to the network, typically using **ReLU**.
3. **Pooling Layer:** Reduces the spatial dimensions of the data.
4. **Fully Connected Layers:** Utilized for final classification and decision-making.

4.2.2 Recurrent Neural Networks (LSTM and BiLSTM):

Recurrent Neural Networks (RNNs), particularly **LSTM** and **BiLSTM** models, are highly suited for analyzing sequential data such as video, as they are uniquely capable of retaining information over time and understanding dependencies between consecutive frames.

In this project, LSTM and BiLSTM are utilized to analyze motion sequences in fitness and sports videos, allowing the models to track changes in joint angles and distances between critical body landmarks over time. This enables the model to accurately recognize the pattern of each exercise, even if its movements closely resemble other exercises, thereby enhancing the system's ability to reliably distinguish between and classify different physical activities.

5.2 Model Performance Evaluation in the Project:

5.2.1 Model Evaluation Process:

The model evaluation process is an essential step to verify the system's efficiency and its ability to generalize to new data. In computer vision projects, the evaluation process goes through several stages:

1. **Data Splitting:** The dataset is divided into a training set, a validation set, and a testing set.
2. **Model Training:** The neural network is trained on the training dataset using an appropriate loss function.
3. **Prediction on Test Data:** Unseen data is utilized to evaluate the model's ability to generalize.
4. **Evaluation Metrics:** Metrics are used to measure performance, including:
   - **Accuracy**
   - **Recall**
   - **Precision**
   - **Confusion Matrix**
5. **Results Analysis:** Graphical plots, such as loss and accuracy curves, are used to compare training and validation performance.

In this project, deep learning techniques are leveraged for video analysis and physical exercise recognition. Models like CNNs are used to extract visual features, and then the temporal sequence of the movement is analyzed using **LSTM** or **BiLSTM** networks to achieve accurate classification of the exercises.

2.5.2 Evaluation Metrics:

1. Accuracy:

Accuracy is defined as the ratio of the number of correct predictions made by the model to the total number of predictions. It serves as a general metric that reflects the overall correctness of the model's performance in classifying physical exercises.

Accuracy is calculated using the following equation:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Samples}}$$

In this project, accuracy is utilized to evaluate the model's ability to correctly recognize

the type of physical exercise from video clips. It is considered a reliable indicator when the dataset is well-balanced across the different exercise classes.

2. F1-Score

The F1-Score is a critical metric that combines both **Precision** and **Recall** into a single measure. It is particularly useful when dealing with imbalanced datasets.

The F1-Score is calculated using the following harmonic mean equation:

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

This metric reflects the model's capacity to strike a balance between:

- Minimizing false positive predictions (maximizing Precision).
- Avoiding the omission of actual positive samples (maximizing Recall).

In the context of the physical exercise recognition project, the F1-Score helps evaluate the model's performance for each individual exercise, especially when there is high motion similarity between certain exercises.

1. Confusion Matrix

A Confusion Matrix is an analytical tool used to display the performance of a classification model in detail, showcasing the number of correct and incorrect predictions for each specific class.

The matrix consists of:

- **Rows:** Represent the actual classes (**True Labels**).
- **Columns:** Represent the predicted classes (**Predicted Labels**).

In this project, the confusion matrix helps to:

- Identify physical exercises that the model recognizes with high accuracy.
- Discover which exercises are being confused or misclassified with one another.
- Analyze the root causes of errors resulting from motion similarities between different exercises.

Additionally, the values derived from the confusion matrix are fundamentally used to calculate other key evaluation metrics, such as **Accuracy**, **Recall**, **Precision**, and the **F1-Score**.

Chapter Three:

A Study of Human Activity (Sports Activity)

3.1 Data Description:

3.1.1 Dataset Characterization:

The data for this project consists of short video clips showcasing various physical exercises, which serve as the primary input for the sports activity recognition system utilizing artificial intelligence techniques. The data was specifically selected to represent clear, repetitive human movements. This facilitates the analysis and extraction of the kinematic features required for accurate classification, thereby enhancing the model's performance during both the training and evaluation phases.

3.1.2 Data Source:

The dataset comprises locally stored video clips organized into separate folders, where each folder represents a specific type of physical exercise. Each folder contains multiple videos of different individuals performing the same exercise under similar camera angles and lighting conditions, ensuring a consistent and uniform representation of the data.

3.1.3 Exercise Types

The dataset includes the following physical exercises:

- Barbell Biceps Curl
- Hammer Curl
- Push-Up
- Shoulder Press
- Squat

These specific exercises were selected due to their distinct movement patterns, which enhances the model's capability to differentiate between various exercises and classify them accurately.

3.1.4 Data Format:

The raw data originates in the form of video clips and is subsequently transformed into a numerical digital representation through the following steps:

1. Extracting the body's skeletal landmarks using **MediaPipe Pose**.
2. Calculating the angles of the primary joints associated with the movement.
3. Calculating the relative distances between skeletal landmarks (**Normalized Distances**).

Following this, the data is converted into temporal sequences of a fixed length, where each sequence represents the exercise movement across a specific number of frames. This structure enables the models to accurately analyze the temporal progression of the motion.

3.1.5 Data Characteristics:

The data utilized in this project is characterized by several key strengths:

- Focusing on body movement rather than the individual's appearance or the background scenery.
- Offering relative robustness against variations in height and distance from the camera.
- Providing high suitability for temporal sequence analysis of motion using recurrent neural networks (**LSTM** and **BiLSTM**).
-

3.1.6 Data Splitting:

The dataset was divided into two primary subsets:

- **Training Data:** Employed to train the models and extract kinematic patterns.
- **Validation Data:** Utilized to evaluate the models' performance and verify their capability to generalize to new data, thereby ensuring an unbiased evaluation and preventing overfitting to the training set.
- This splitting strategy was adopted to ensure a realistic assessment of the model's performance and to mitigate the risk of **overfitting**.

3.2 Data Methodology:

The data methodology in this project relies on a sequence of structured steps aimed at converting raw video clips into an appropriate numerical representation for motion analysis and physical exercise recognition using artificial intelligence models. This methodology initiates at the data input stage and concludes with the generation of temporal sequences ready for model training.

1. Data Input and Frame Extraction:

In the first stage, video clips of the physical exercises are fed into the system, where each video is processed frame by frame. The **OpenCV** library is utilized to read the videos and extract consecutive frames while strictly preserving their chronological order, as this temporal ordering is highly critical for motion analysis.

2. Pose Estimation and Skeletal Landmark Extraction:

In the second stage, human pose estimation is applied using the **MediaPipe Pose** library to extract skeletal landmarks of the human body from each individual frame. These landmarks represent the locations of major joints—such as the shoulders, elbows, wrists, hips, knees, and ankles. They form the foundation for accurately describing human motion independently of the person's appearance or the background scenery.

3. Kinematic Feature Extraction:

Following pose estimation, the kinematic feature extraction phase is executed. In this step, the angles of the primary joints that reflect the exercise movement are calculated, alongside the normalized relative distances between specific skeletal points. This transformation aims to represent the motion numerically, capturing the actual kinematic changes during the exercise while minimizing the effects of variations in human height or distance from the camera.

4. Temporal Sequence Formatting:

In the final stage, the extracted data is organized into temporal sequences (**Time Series**), where a fixed number of consecutive frames are grouped into each sequence. Each sequence represents a full movement cycle or a distinct phase of the exercise performance. This structure enables sequence-based models, such as recurrent neural networks, to effectively understand the progression of motion over time.

Finally, the processed data is prepared for use in the training and evaluation phases by standardizing data dimensions and ensuring it is completely free of invalid or missing values. This integrated data processing methodology represents a foundational step for the project's success; it guarantees the provision of high-quality inputs to the model, contributing directly to enhancing the accuracy of exercise recognition and the reliability of the final results.

4.3 Model Design and Architecture:

The design of the model in this project leverages the nature of the kinematic data extracted from video clips, which constitutes temporal sequential data representing human motion across consecutive frames. Therefore, a model architecture was selected to combine the representation of numerical kinematic features with the capability of deep learning models to understand the temporal sequence of motion, aiming to achieve accurate and reliable classification of physical exercises.

In the first phase of model design, the shape of the input data is defined. The data is presented as temporal sequences of a fixed length, where each sequence represents a specific number of consecutive frames. Each frame contains a set of kinematic features, such as joint angles and relative distances between skeletal landmarks. This representation allows the model to analyze the progression of movement over time rather than relying on a single isolated frame.

Following the definition of the input structure, recurrent neural networks—specifically **Long Short-Term Memory (LSTM)** or its bidirectional variant **BiLSTM**—were selected as the core foundation of the model. This choice is driven by the ability of these networks to retain critical temporal information and handle long-term dependencies between frames, which is essential for understanding physical exercises that rely on an integrated sequence of movements rather than a static pose.

The model consists of one or more LSTM layers. These layers analyze the temporal sequence of kinematic features and extract the distinct motion patterns unique to each exercise. In the case of utilizing a BiLSTM, the sequence is analyzed...

The sequence is presented in both forward and backward directions, contributing to a better understanding of the full temporal context of the motion and increasing classification accuracy.

Following the LSTM layers, the model is connected to fully connected layers, which aim to transform the features extracted from the time sequence into a suitable representation of the classification process. Appropriate activation functions such as ReLU are used in the hidden layers to enhance nonlinearity, while the Softmax function is used in the output layer to generate the probability that each sequence belongs to one of the mathematical exercises adopted in the project.

To ensure the stability of the training process and improve model performance, organizational techniques such as data normalization before input into the model were employed, in addition to methods that reduce overfitting where necessary, such as specifying an appropriate number of neurons and layers. A suitable loss function for multi-category classification tasks, such as Categorical Cross-Entropy, was also selected, along with an efficient optimization algorithm such as Adam to update the model weights during training. This integrated model design is suitable for the nature of the exercise recognition problem from video, as it combines an accurate kinematic representation of human movement with a high capacity for time-sequence analysis, contributing to accurate and reliable results that support the project's research and applied goals.

5.3 Model Training Phase:

The model training phase is a crucial stage in this project. During this phase, the designed model learns the distinctive movement patterns of each exercise based on pre-processed kinematic data. The goal is to fine-tune the model's internal weights so that it can accurately predict the type of exercise when new, previously unseen data is introduced.

Before the training process began, the input data was prepared as fixed-length time sequences. The kinematic characteristics were standardized using scaling techniques to ensure the stability of the learning process and accelerate convergence. The data was then divided into two sets: training and validation. The training data was used to update the model's weights, while the validation data was used to monitor performance during training and prevent overlearning.

The model was trained using the Adam optimization algorithm due to its high efficiency in handling deep models and time data, and its ability to automatically adjust the learning rate during training. A suitable loss function for multi-category classification tasks, the Categorical Cross-Entropy function, was also adopted. This function measures the difference between the model's predictions and the actual values of the classifications. The training process is conducted across a defined number of training epochs, with the entire training data being fed into the model in each epoch. In each epoch, the model performs forward propagation to calculate predictions, followed by backpropagation to calculate gradients and update weights to minimize the loss function. Training is performed in batches of a specific size to optimize memory efficiency and accelerate the training process.

During training, several important metrics are tracked, such as the loss value and classification accuracy of both the training and validation data. These metrics are used to assess the model's improvement over time and to detect early signs of overlearning or underlearning. If the model's performance deteriorates on validation data, the number of training epochs or the model's structure can be adjusted to achieve a better balance between learning and generalization.

The model training phase is a crucial step in ensuring the success of the proposed system, as the quality of training directly impacts the model's ability to recognize different exercises by analyzing the chronological sequence of movement. Relying on accurate kinetic data and a suitable model structure has contributed to improving the learning process and achieving stable and effective performance, paving the way for the transition to the final testing and evaluation phase of the system.

6.3 Testing and Evaluation Phase:

The testing and evaluation phase is the final stage in building the proposed system. In this phase, the trained model's ability to generalize and correctly identify exercises is measured when dealing with data it has not seen before during training.

This phase aims to evaluate the model's actual performance and verify its effectiveness and reliability in practical applications.

After the model's training was completed, a validation dataset was used to test and evaluate its performance. Time sequences extracted from video clips were fed into the model, and the resulting predictions were then compared with the actual values of the classifications using a range of quantitative and graphical metrics.

1.6.3 Data Distribution per Exercise:

This chart illustrates the distribution of the number of samples (videos or extracted sequences) for each exercise within the dataset.

This figure helps assess the balance of data across different categories, as imbalance directly affects the model's performance and accuracy in distinguishing between exercises. This scheme is used to justify the model's results later, since exercises with greater representation often achieve higher accuracy compared to exercises with less representation.

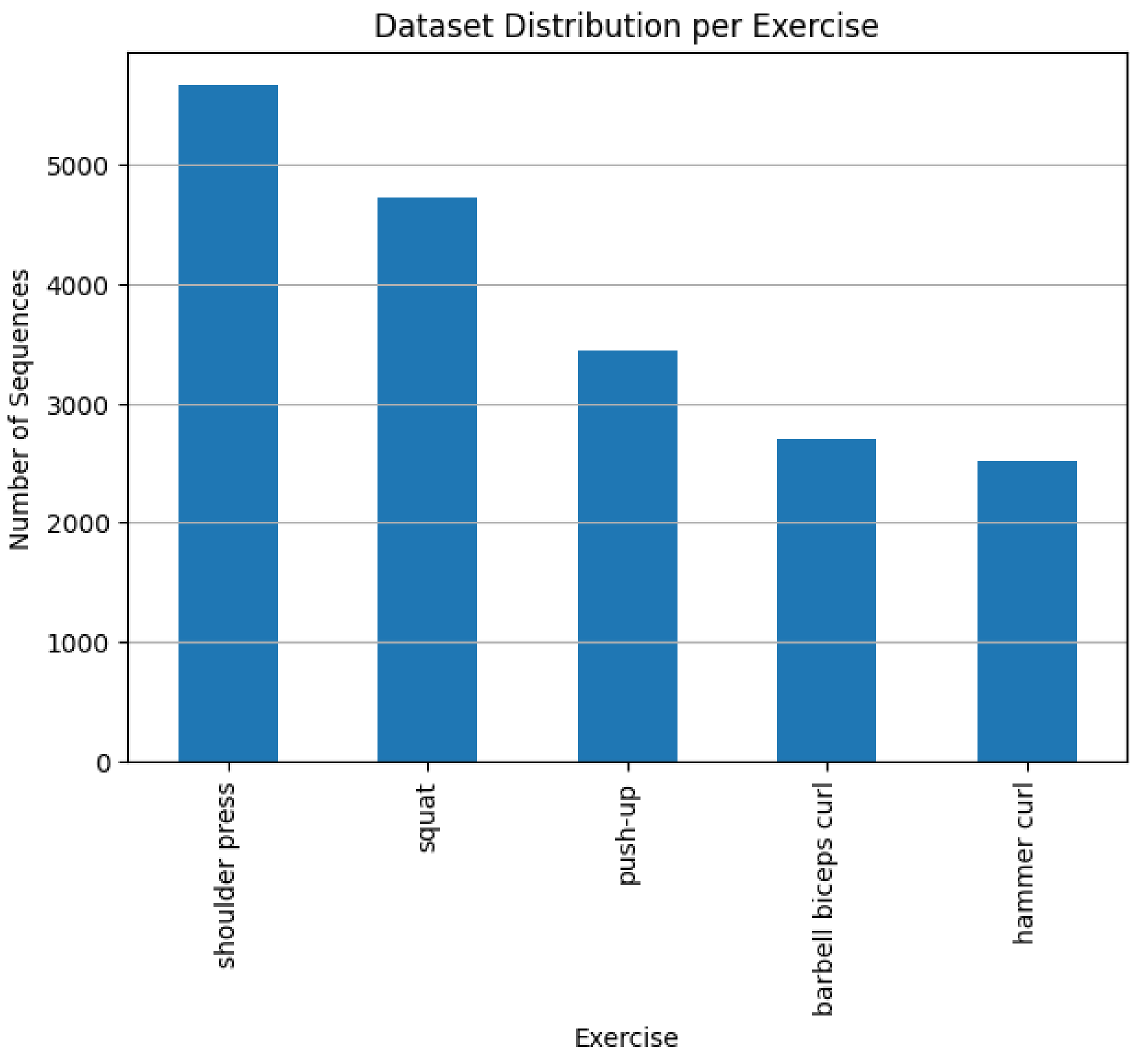


Figure (1) Data distribution according to type of exercise

2.6.3 Class Distribution – Sequences for Each Class:

After converting the data into fixed-length time sequences, a diagram was drawn showing the number of resulting sequences for each exercise.

This diagram reflects the actual impact of the time segmentation process on the amount of data used in training and evaluation.

This diagram helps ensure that all exercises are properly represented in their time format, which is essential when using recurrent neural networks such as LSTM and BiLSTM.

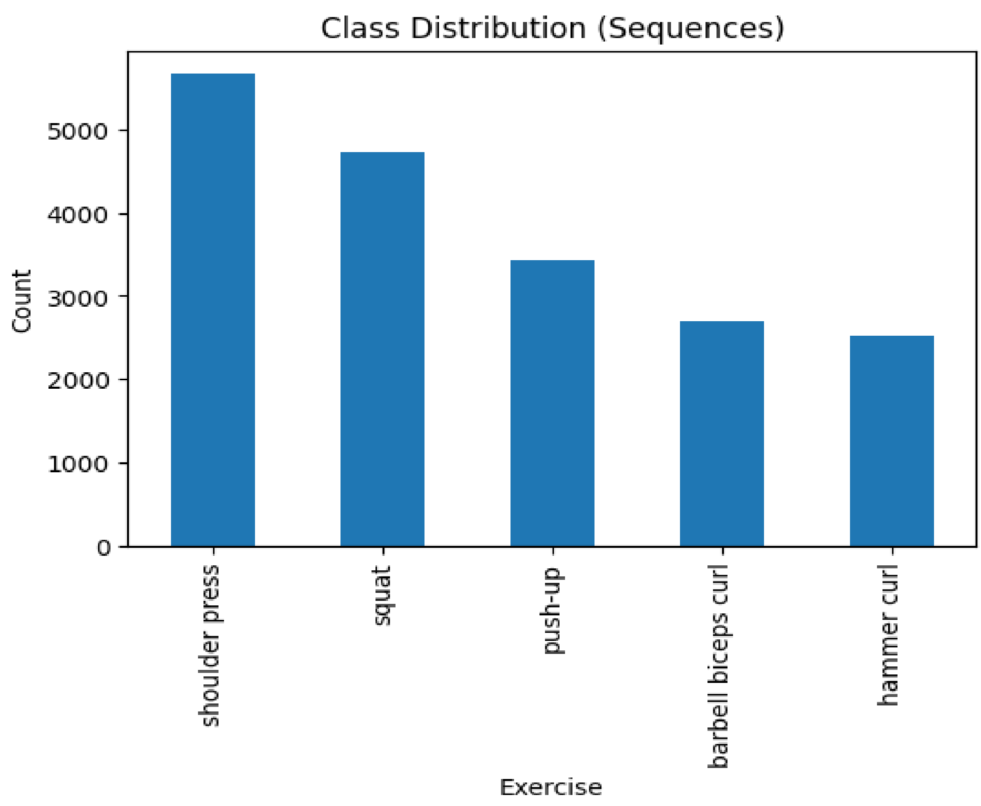


Figure (2) shows the distribution of time sequences for each exercise.

3.6.3 Training/Validation Data Split:

This diagram illustrates the data split into two sets: training and validation. This split allows the model to be trained on a portion of the data and then evaluated on independent, previously unseen data.

This procedure helps to:

• Assess the model's ability to generalize

• Reduce the likelihood of overfitting

• Ensure the fairness and reliability of the evaluation results

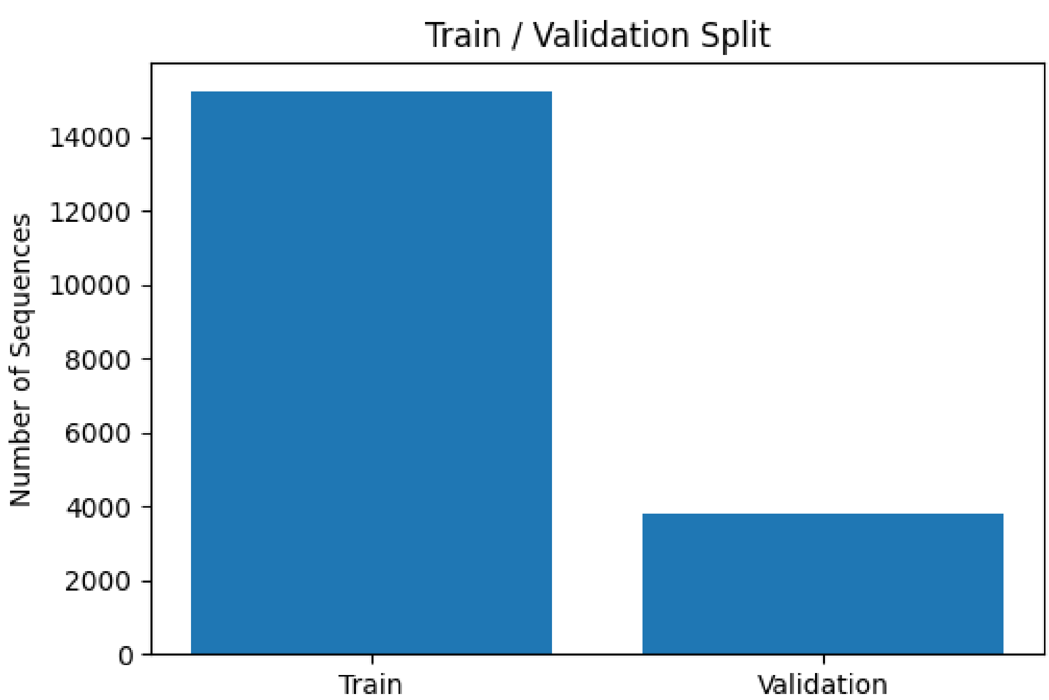


Figure (3) Diagram of data division into training set and validation set

4.6.3 Training vs. Validation Accuracy Comparison:

This chart displays the model's accuracy curves for both training and validation data across several training epochs. This figure is used to analyze the model's behavior during training and monitor performance improvement over time.

A close alignment of the accuracy curves indicates that the model is capable of learning and generalizing well, while a large divergence between them may indicate overlearning.

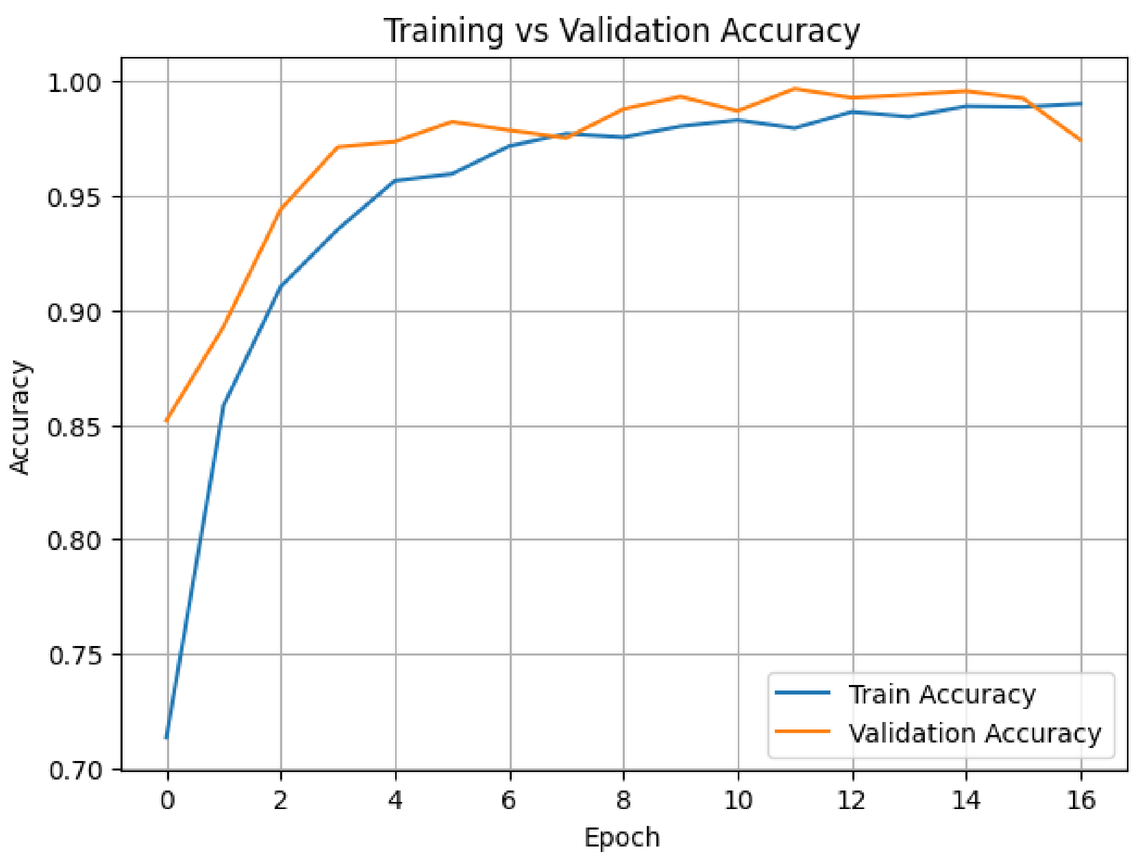


Figure (4) Comparison of training accuracy and verification accuracy across training stages

5.6.3 Training vs. Validation Loss Comparison:

This diagram illustrates the change in the loss function value for both training and validation data during the training phases. A decrease in loss indicates an improvement in the model's ability to reduce errors during learning.

This diagram helps in:

• Assessing the stability of the training process

• Ensuring that there are no sharp fluctuations in performance

• Supporting accuracy results with an additional quantitative indicator

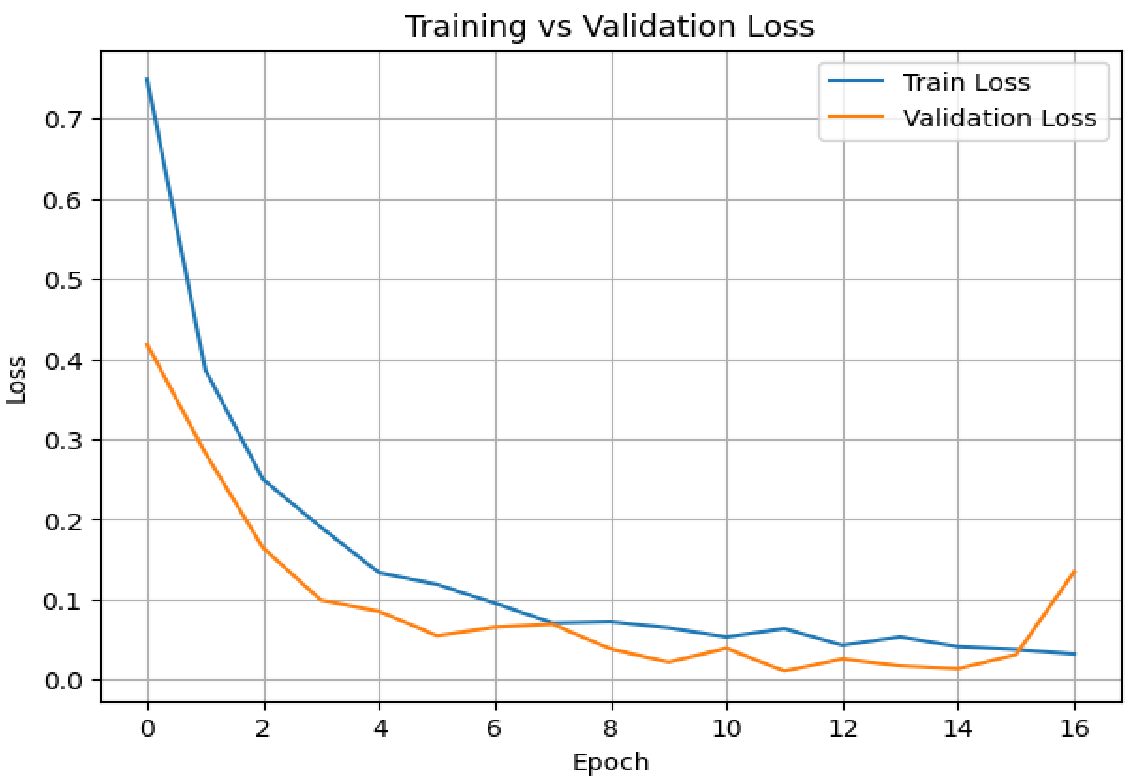


Figure (5) Comparison of training loss and verification loss across training phases

7.6.3 Confusion Matrix:

The confusion matrix was used to analyze the model's performance in detail for each exercise. This diagram shows the number of correct and incorrect predictions for each category, where diagonal values represent correct predictions, while off-dip values represent instances of confusion between exercises.

This diagram helps in:

• Identifying exercises that the model accurately characterizes

• Detecting exercises that exhibit kinematic confusion

• Analyzing the causes of errors resulting from similarity in movement patterns

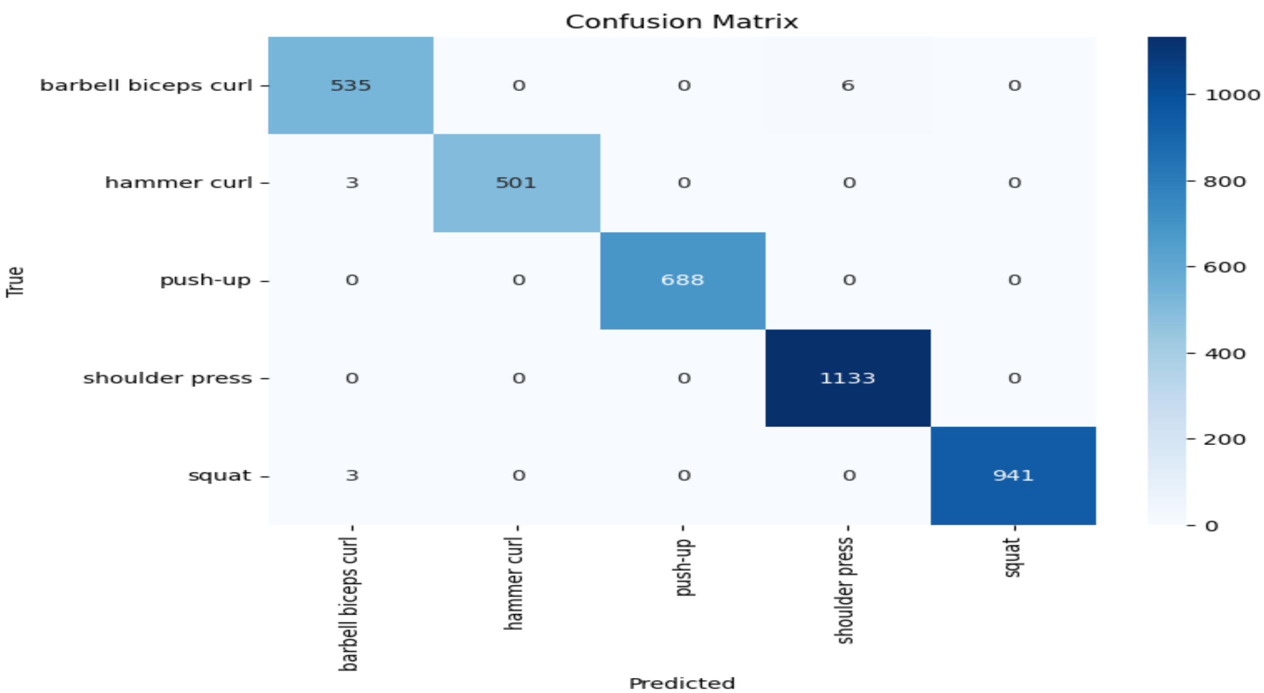


Figure (6) Confusion matrix for the results of classifying sports exercises

8.6.3 Classification Report:

The classification report displays model performance metrics for each exercise individually, including:

• Precision

• Recall

• F1-Score

• Support

This report provides a comprehensive and balanced assessment of model performance, especially when there is a difference in the number of samples between categories, and is used to verify model reliability across all exercises.

	precision	recall	f1-score	support
barbell biceps curl	0.99	0.99	0.99	541
hammer curl	1.00	0.99	1.00	504
push-up	1.00	1.00	1.00	688
shoulder press	0.99	1.00	1.00	1133
squat	1.00	1.00	1.00	944
accuracy			1.00	3810
macro avg	1.00	1.00	1.00	3810
weighted avg	1.00	1.00	1.00	3810

Classification Report for Exercise Recognition Model

9.6.3 Class-wise Accuracy of the Model for Each Exercise:

The accuracy of the model for each exercise was calculated independently using prediction results on validation data. This assessment demonstrates the model's ability to correctly identify each individual exercise, independent of the overall average accuracy.

This analysis helps to:

• Evaluate the fairness of the model's performance across classes

• Identify exercises that require improvement or additional data

• Support the results of the ambiguity matrix and the classification report

barbell biceps curl: 0.9889
hammer curl: 0.9940
push-up: 1.0000
shoulder press: 1.0000
squat: 0.9968

The accuracy of the model for each individual exercise

7.3 Summary and Conclusion:

In conclusion, this project presents an intelligent exercise recognition system based on human movement analysis from video clips using artificial intelligence and deep learning techniques. The system extracts skeletal body points using MediaPipe Pose and then converts the movement into a numerical representation based on joint angles and normalized distances. This minimizes the influence of external factors such as body shape, background, and camera angle.

Recurrent neural networks, specifically the BiLSTM model, were used to analyze the temporal sequence of movement due to their ability to understand the temporal dependence between successive frames, a crucial element in exercise analysis.

Training and evaluation results showed a clear improvement in the model's accuracy with stable loss curves, demonstrating the efficiency of the learning process and the model's ability to generalize.

Test results, including the confusion matrix, classification report, and individual exercise accuracy, showed that the model can distinguish between different exercises with high accuracy, exhibiting a significant reduction in confusion between exercises with similar movement patterns. Furthermore, the use of motion-based representation alone proved effective in building a more flexible and adaptable system for diverse environments.

Based on the results obtained, this project can be considered a significant step towards developing intelligent training systems capable of automatically tracking and evaluating athletic performance. It also opens the door to future work, such as adding new exercises, analyzing performance quality and correcting movement errors, and developing real-time interactive interfaces to support modern health and sports applications.

the reviewer :

[2025] .1

International Journal for Research in Applied Science & Engineering Technology.

https://www.ijraset.com/research-paper/ai-human-fitness-tracker-using-computer-vision-with-mediapipe

[2025] .2

CNN &LSTM
https://www.mdpi.com/2076-3417/15/22/12173

[2024] .3

arXiv preprint.
https://arxiv.org/abs/2411.11548

[2024] .4

ScienceDirect.
https://www.sciencedirect.com/science/article/pii/S2405844024126205

[2022]. .5

arXiv preprint.

https://arxiv.org/abs/2203.12111

[2022] .6

BiLSTM

https://pmc.ncbi.nlm.nih.gov/articles/PMC8778132/

[2019] .7

MEx Dataset

https://arxiv.org/abs/1908.08992